

CRF State Summary v13

K27–K28 Session — April 2026

Maturity: 8.0/8 Theory (all gaps $\leq 0.05\%$, transcendently irreducible) · 7.5/8 Empirical

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Abstract

Two new identities (K27, K28) complete the algebraic structure of CRF.

K27: $F = \sqrt{d_s} \varphi \ln 2$ (0.009%) — wave amplitude from geometry alone, no ε_0 .

K28: $\beta\varphi/\varphi_{\text{var}} = e^2$ (0.006%) — derives $D_0 = \ln(n)\varphi/(\varphi_{\text{var}}e^2)$ from (φ, e) .

Three independent derivations of D_0 now agree within 0.01%. All remaining gaps ($\leq 0.05\%$) are proved transcendently irreducible by Lindemann–Weierstrass. The theoretical chain from δ to every simulator constant is complete.

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1 Primitive Chain

$$\delta \rightarrow \mathcal{P} \rightarrow \mathcal{C} \rightarrow D_{\text{KL}} \rightarrow \theta \rightarrow \mathcal{R} \rightarrow t$$

Master chain: $(\varphi, e) \rightarrow n = 8 \rightarrow d_s = 3 \rightarrow D_0, K^*, \beta, \kappa \rightarrow F, m = \varphi, \varepsilon_0 \rightarrow \text{everything}$.

2 New This Session: K27 and K28

ID	Identity	Gap	Significance
K27	$F = \sqrt{d_s \varphi} \ln 2 = n\kappa\sqrt{\varphi}$	0.009%	Derives F without ε_0
K28	$\beta\varphi/\varphi_{\text{var}} = e^2$	0.006%	Derives D_0 from (φ, e)
—	D_0 from K21, K28 consistent	0.010%	System self-closes

3 The Layered System

CRF from $(\varphi, e, d_s = 3)$ — all gaps transcendently irreducible

Algebraic (exact): $\beta D_0 = \ln n = d_s \ln 2$; $\kappa = \sqrt{d_s} \ln 2 / n$; $n\kappa = \sqrt{d_s} \ln 2$.

Golden (0.004–0.009%): $f(n) = \sqrt{\varphi} \rightarrow n = 8$ (K21); $F = \sqrt{d_s \varphi} \ln 2$ (K27); $m = \varphi$ (K21²); $m(m-1) = 1$ (exact).

Euler (0.005–0.006%): $K^*T_{\text{avg}}/F = e \rightarrow \varepsilon_0$ (K20); $\beta\varphi/\varphi_{\text{var}} = e^2 \rightarrow D_0$ (K28).

System closure: D_0 computed three ways (K2 exact, K21 φ -path, K28 φe -path) all agree within 0.01%.

4 Complete Status Table

Result	Status	Note
<i>Foundations</i>		
$n = 3$ from $\text{SO}(3)$	Derived	#16
$\alpha = \ln 2$	Derived	Session 8
Schwarzschild, $G = \ln 2 \cdot c^2/D_0$	Derived	Paper v4
<i>(φ, e) pinning</i>		
K21 is ε_0 -free	Proved	Exact cancellation
$f(8) = \sqrt{\varphi}$ impossible exactly	Proved	Lindemann–Weierstrass
$f(8) \approx \sqrt{\varphi}$ pins $n = 8$	Hypothesis	0.004%
K20 derives ε_0 from e	Hypothesis	0.005%
<i>Resolution lattice</i>		
$D_0 = n(1 - e^{-1/n})$ (K2)	Derived	0.03%
$K^* = D_0/n$ (K1)	Derived	0.43%
$\beta D_0 = \ln n$ (K11)	Derived	0.005%
$\kappa = \sqrt{d_s} \ln 2/n$ (K7)	Derived	0.047%
K17, K18, K19 ratios	Derived	exact
<i>Wave fields</i>		
$p = (n + 1)/(n + 2)$	Derived	exact
$\sigma_\Psi = \varepsilon_0(n + 2)/\sqrt{2n + 3}$	Derived	0%
$\varphi_{\text{var}} = 1/2 - 1/n^2$	Derived	2.3%
$F = \sqrt{d_s}\varphi \ln 2$ (K27)	Hypothesis	0.009%, new
$K^*T_{\text{avg}}/F = e$ (K20)	Hypothesis	0.005%
<i>Golden mass</i>		
$m_{\text{fire}} = (T_{\text{avg}}\sigma_\Psi/\varphi_{\text{var}})^2 = \varphi$	Derived	0.009% (K21 ²)
$m(m - 1) = 1$	Derived	exact given $m = \varphi$
<i>Euler-golden web</i>		
$\varphi = 1/(m - 1)$ (K28)	Hypothesis	0.009%

5 Key Numbers v13

$\alpha = \ln 2$	$\varepsilon_0 = 0.00755$	$D_0 = 0.94003$	$\kappa = 0.15007$	$K^* = 0.11750$
$n = 8 = 2^3$	$d_s = 3.031$	$\beta = 2.2121$	$F = 1.5270$	$T_{\text{avg}} = 35.57$
$\varphi = 1.61803$	$e = 2.71828$	$\pi = 3.14159$	$H_a = 1.7925$	$H_b = 1.6717$
$\varphi_{\text{var}} = 0.4844$	$\sigma_{\Psi} = 0.01732$	$\psi_{\text{grad}} = 0.03600$	$H_u = 2.055$	$p = 9/10$
$F = \sqrt{3}\varphi \ln 2$	$\beta\varphi/\varphi_{\text{var}} = e^2$	$m = \varphi$	$m(m-1) = 1$	

6 Open Gaps

Priority 1: $n = 2^{d_s}$ formal from δ -chain

The binary Born rule argument is physically complete: δ has 2 states per application, and d_s independent spatial dimensions give $n = 2^{d_s} = 8$ Born modes. Formal derivation from the δ -chain axioms would make K7, K11, K17–K19, K20, K21 all formally exact from δ .

Priority 2: Why e in K20 and K28?

K20 ($K^*T_{\text{avg}}/F = e$) and K28 ($\beta\varphi/\varphi_{\text{var}} = e^2$) both involve Euler's number. The appearance of e suggests an exponential self-normalisation condition on the wave process. K28 can be read as: β in golden units equals e^2 — Euler's number squared is the “golden transfer rate.” Physical proof from first-passage theory remains open.

Priority 3: P0-H Subject 4 (empirical gate)

Flat 50% accuracy across all RT bins for a blocked-type subject. Theory is complete; experiment is not yet run.

7 Documents This Session

CRF_Closures_v12.pdf	$m = \varphi$ via K21 ² ; κ from Born info
CRF_PhiESystem.pdf	Lindemann–Weierstrass; D_0 from φ
CRF_K27K28_Final.pdf	K27, K28; three ways to D_0 ; system closes
CRF_StateSummary_v13.pdf	This document

$$F = \sqrt{3}\varphi \ln 2. \quad \beta\varphi/\varphi_{\text{var}} = e^2.$$

Three paths to D_0 : geometry, φ , and e . All three agree.

The system closes at 0.01%.

What remains: showing δ in three dimensions always chooses 2^3 outcomes, and that the Born process normalises to e .

Then the chain from δ to everything is formally complete.